

## Hepatic encephalopathy

### Altered consciousness

- **Altered consciousness:** A state where a person's awareness or alertness is reduced or altered, ranging from confusion to a deep coma.

### Asterixis (flapping tremor)

- **Asterixis:** A type of involuntary, rapid, and irregular movement (often seen in the hands) that resembles flapping.

### Pathophysiology

- **Ammonia:** A toxic substance produced in the body, typically by bacteria in the intestines, which can affect the brain if the liver cannot remove it properly.

### Diagnosis

- **Electroencephalogram (EEG):** A test that measures electrical activity in the brain, which may show irregularities in cases of encephalopathy.

### Management

- **Lactulose:** A medication that helps lower ammonia levels in the body by drawing it into the gut to be excreted.
- **Rifaximin:** An antibiotic that targets bacteria in the gut, reducing the production of harmful toxins.

### Prognosis

- **Prognosis:** The likely outcome or course of a disease or condition, including chances of recovery or complications.

## Liver abscess

### Portal vein bacteraemia

- **Bacteraemia:** The presence of bacteria in the bloodstream, which can spread infection to various organs.



## Ascending cholangitis

- **Ascending cholangitis:** A bacterial infection of the bile ducts that moves upwards from the intestines to the liver.

## Subphrenic abscess

- **Subphrenic abscess:** A collection of pus located under the diaphragm, often as a result of infection in nearby organs.

## Hepatic haematoma

- **Hepatic haematoma:** A collection of blood in the liver, often due to trauma, which may later become infected.

## Anaerobic bacteria

- **Anaerobic bacteria:** Bacteria that can survive and grow in environments without oxygen.

## Rigors

- **Rigors:** Shivering or shaking due to fever, often accompanied by chills.

## Atelectasis

- **Atelectasis:** Collapse or incomplete expansion of the lung tissue, leading to reduced oxygen supply.

## Pus aspiration

- **Pus aspiration:** The process of removing pus (a thick fluid filled with white blood cells and bacteria) from an abscess or infected area using a needle.

## Sensitivity reports

- **Sensitivity reports:** Lab tests that identify which antibiotics are most effective against a particular bacterial infection.



## Ultrasound-guided aspiration

- **Ultrasound-guided aspiration:** A procedure where an ultrasound is used to guide a needle to a specific location to remove fluid or pus for testing.

## Hepatitis B

- **DNA virus:** A virus that uses DNA as its genetic material for replication.
- **DNA polymerase enzyme:** An enzyme that helps replicate DNA, crucial for virus reproduction.

## Route of Transmission

- **Horizontal transmission:** Spread of infection from one individual to another through direct or indirect contact (e.g., through blood or sexual contact).
- **Vertical transmission:** Transmission of infection from an infected mother to her baby during childbirth.

## High-Risk Groups

- **Haemodialysis:** A medical treatment for kidney failure that filters waste from the blood, often exposing individuals to potential infections.
- **Paramedical staff:** Healthcare workers who assist doctors, such as nurses or laboratory technicians.

## Acute Hepatitis Clinical Features

- **Prodromal phase:** The early stage of infection, before the appearance of the characteristic symptoms.
- **Anorexia:** Loss of appetite.
- **Malaise:** General feeling of discomfort or illness.
- **Myalgia:** Muscle pain.
- **Arthralgia:** Joint pain.
- **Icteric phase:** The stage where jaundice (yellowing of skin and eyes) appears.
- **Jaundice:** A condition where the skin and the whites of the eyes turn yellow due to high bilirubin levels.
- **Cholestasis:** A reduction or stoppage of bile flow, often leading to jaundice and pruritus.
- **Anicteric hepatitis:** A form of hepatitis where jaundice does not occur.



## Signs of Acute Hepatitis

- **Yellow sclera:** Yellowing of the white part of the eyes, often a sign of jaundice.
- **Pruritus:** Itching of the skin, often due to the buildup of bile salts in cholestasis.
- **Hepatomegaly:** Enlarged liver.
- **Splenomegaly:** Enlarged spleen.
- **Lymphadenopathy:** Swollen lymph nodes.

## Complications of Acute Hepatitis

- **Acute fulminant hepatitis:** A severe, rapid onset form of liver failure that can lead to death.
- **Relapsing hepatitis:** Recurrence of hepatitis symptoms after initial improvement.
- **Cholestatic hepatitis:** Hepatitis with impaired bile flow, leading to jaundice and itching.
- **Aplastic anaemia:** A condition where the bone marrow stops producing enough blood cells, leading to low red blood cell count.
- **Papular acrodermatitis:** A skin condition characterized by small, raised bumps on the skin, often linked to hepatitis B infection.
- **Myelitis:** Inflammation of the spinal cord.
- **Neuropathy:** Nerve damage, which can cause pain, tingling, or loss of function.

## Investigations

- **Serology:** Blood tests that detect antibodies or antigens related to a virus.

## HBV Antigens

- **HBsAg (Hepatitis B Surface Antigen):** A protein on the surface of the hepatitis B virus that is used to detect infection.
- **HBcAg (Core Antigen):** A protein found in the core of the hepatitis B virus, not detectable in blood.
- **HBeAg (Hepatitis E Antigen):** A protein that indicates active viral replication and high infectivity.

## Viral Blood

- **HBV-DNA:** Genetic material of the hepatitis B virus, detectable in the blood by PCR (Polymerase Chain Reaction) testing.
- **Viral load:** The quantity of virus present in the blood, with higher levels indicating active replication.



## Other Investigations

- **ALT (Alanine Aminotransferase):** An enzyme found in the liver that increases when the liver is damaged.
- **AST (Aspartate Aminotransferase):** Another enzyme that rises in response to liver injury.
- **Alkaline phosphatase:** An enzyme that increases in cases of cholestasis (impaired bile flow).
- **Prothrombin time (PT):** A blood test that measures how long it takes for blood to clot; prolonged PT indicates severe liver damage.

## Management

- **Supportive treatment:** Care that focuses on relieving symptoms and preventing complications rather than curing the disease.
- **Hepatotoxic drugs:** Medications that can damage the liver.
- **Cholestyramine:** A medication used to relieve itching in cases of cholestasis.
- **Subcutaneous interferon alpha:** A treatment for acute hepatitis C that is given by injection.

## Prevention of Viral Hepatitis

- **Prophylaxis:** Preventative treatment or measures to reduce the risk of infection.
- **Active immunization:** Vaccination to protect against viral infections before exposure.
- **Recombinant hepatitis B vaccine:** A vaccine made using recombinant DNA technology that helps the body develop immunity to hepatitis B.

## Hepatocellular injury

- **Hepatocellular injury:** Damage to liver cells, which can lead to scarring and dysfunction.

## Aetiology

- **Autoimmune hepatitis:** A condition where the immune system attacks liver cells, leading to inflammation and damage.
- **Drug-induced:** Liver damage caused by medications or drugs.
- **Biliary cirrhosis:** Chronic liver disease caused by damage to the bile ducts, leading to scarring.
- **Hemochromatosis:** A genetic disorder causing excessive iron buildup in the body, damaging the liver.



- **Wilson's disease:** A genetic disorder that leads to copper accumulation in the liver and other organs.
- **Cardiac cirrhosis:** Cirrhosis caused by chronic heart failure leading to blood congestion in the liver.
- **Alpha-1 antitrypsin deficiency:** A genetic disorder that can cause liver and lung damage due to insufficient production of a protective enzyme.
- **Idiopathic:** A condition without a known cause.

## Pathogenesis

- **Stellate cells:** Cells in the liver that, when activated, produce collagen and contribute to fibrosis (scar tissue formation).
- **Fibrosis:** The thickening and scarring of tissue, often due to chronic injury.
- **Micronodular cirrhosis:** Cirrhosis with small regenerative liver nodules, often seen in alcoholics.
- **Macronodular cirrhosis:** Cirrhosis with larger nodules, seen in conditions like post-hepatitic cirrhosis.

## Clinical Features

- **Portal hypertension:** Increased blood pressure in the portal vein, which can cause complications like ascites.
- **Ascites:** The accumulation of fluid in the abdomen, often due to liver disease.
- **Splenomegaly:** Enlargement of the spleen.
- **Haematemesis:** Vomiting of blood, often due to bleeding from varices.
- **Melaena:** Black, tarry stools due to gastrointestinal bleeding.
- **Hepatic encephalopathy:** Brain dysfunction caused by liver failure, leading to confusion and altered mental status.
- **Oedema:** Swelling due to fluid retention, often in the legs or abdomen.
- **Coagulopathy:** A blood clotting disorder caused by liver dysfunction.
- **Spider nevi:** Small, spider-like blood vessels visible under the skin.
- **Palmar erythema:** Redness of the palms, often associated with liver disease.
- **Gynaecomastia:** Enlargement of breast tissue in men, often related to liver dysfunction.
- **Testicular atrophy:** Shrinking of the testicles, which can occur in liver disease.
- **Loss of libido:** Reduced sexual desire, common in liver disease.
- **Menstrual abnormalities:** Irregular periods or absence of menstruation in females.
- **Dupuytren's contracture:** Thickening of tissue in the palm of the hand, leading to bent fingers.
- **Skin pigmentation:** Darkening of the skin, often due to liver disease.



## Complications

- **Spontaneous bacterial peritonitis (SBP):** A bacterial infection of ascitic fluid in the abdomen, often a complication of cirrhosis.
- **Hepatorenal syndrome:** A type of kidney failure that occurs in individuals with severe liver disease.
- **Hepatocellular carcinoma:** Liver cancer, which can develop in cirrhotic livers.

## Laboratory Findings

- **Anaemia:** A deficiency of red blood cells, leading to fatigue and weakness.
- **Leukopenia:** A low white blood cell count, making individuals more susceptible to infection.
- **Thrombocytopenia:** A low platelet count, leading to increased bleeding risk.
- **ALT (Alanine Aminotransferase):** An enzyme released when liver cells are damaged, often elevated in liver disease.
- **AST (Aspartate Aminotransferase):** Another enzyme elevated in liver damage.
- **ALP (Alkaline Phosphatase):** An enzyme elevated in bile duct obstruction or cholestasis.
- **Serum albumin:** A protein made by the liver; low levels can indicate liver dysfunction.
- **Prothrombin time (PT):** A blood test that measures clotting ability; prolonged PT suggests liver damage.

## Management

- **Hepatotoxic drugs:** Medications that can harm the liver, should be avoided in liver disease.
- **Vitamin supplementation:** Providing vitamins to support the body's needs, particularly those affected by liver disease.
- **Salt restriction:** Limiting salt intake to reduce fluid retention and ascites.
- **Liver transplantation:** A surgical procedure to replace a damaged liver with a donor liver, often needed for advanced cirrhosis.

## Prognosis

- **Poor prognosis:** The outcome of cirrhosis is often severe, with potential for liver failure, but treatment of the underlying cause can improve the outlook.



## Jaundice

- **Jaundice:** Yellowish discoloration of the skin, mucous membranes, and sclera (white part of the eyes) caused by elevated levels of bilirubin in the blood.
- **Sclera:** The white outer layer of the eyeball.
- **Bilirubin:** A yellow pigment formed when red blood cells break down.
- **Carotenaemia:** The condition where excess carotene in the blood causes yellowing of the skin, but not the sclera.

## Classification of Jaundice

- **Hyperbilirubinaemia:** An elevated level of bilirubin in the blood.
- **Unconjugated hyperbilirubinaemia:** High levels of unconjugated bilirubin in the blood, typically due to overproduction or issues with liver processing.
- **Conjugated hyperbilirubinaemia:** Elevated levels of conjugated bilirubin, often due to liver dysfunction or obstruction of bile flow.

## Unconjugated Hyperbilirubinaemia

- **Overproduction of Bilirubin (Prehepatic):** Increased breakdown of red blood cells, leading to more bilirubin than the liver can process.
- **Hemolytic disorders:** Conditions where red blood cells are destroyed prematurely, causing elevated bilirubin levels.
- **Ineffective erythropoiesis:** The production of abnormal or insufficient red blood cells.
- **Impaired hepatic metabolism:** Reduced ability of the liver to process or conjugate bilirubin.
- **Gilbert's syndrome:** A mild genetic disorder affecting bilirubin processing, leading to intermittent jaundice.
- **Crigler-Najjar syndrome:** A rare genetic disorder that causes severe jaundice due to complete or partial lack of the enzyme needed for bilirubin processing.
- **Kernicterus:** A form of brain damage in neonates caused by very high levels of unconjugated bilirubin.

## Conjugated Hyperbilirubinaemia

- **Dubin-Johnson syndrome:** A rare genetic disorder leading to mild jaundice without liver enzyme abnormalities.
- **Rotor syndrome:** Another genetic disorder similar to Dubin-Johnson syndrome, causing jaundice without liver enzyme issues.
- **Hepatocellular diseases:** Liver diseases that affect its ability to transport bilirubin into bile.



- **Viral hepatitis:** Inflammation of the liver caused by various viruses, such as hepatitis A, B, C, D, E, Epstein-Barr virus (EBV), and cytomegalovirus (CMV).
- **Toxins:** Harmful substances like vinyl chloride (used in plastics) and Amanita phalloides (a type of poisonous mushroom) that can damage the liver.
- **Metabolic diseases:** Conditions like Wilson's disease, where abnormal copper buildup damages the liver.
- **Immune diseases:** Conditions like autoimmune hepatitis where the body's immune system attacks its liver cells.
- **Cholestatic jaundice (Obstructive jaundice):** Jaundice caused by blockage of bile flow, either inside the liver (intrahepatic) or outside the liver (extrahepatic).
- **Intrahepatic causes:** Obstruction of bile ducts within the liver due to conditions like hepatitis, primary biliary cirrhosis, or drug reactions.
- **Extrahepatic causes:** Blockage of bile ducts outside the liver due to gallstones, tumors, or enlarged lymph nodes compressing bile ducts.
- **Stercobilinogen:** A substance formed from bilirubin that contributes to the normal brown color of stools; its absence causes pale stools.
- **Pruritus:** Itching of the skin, commonly caused by bile salts accumulating in the skin.
- **Malabsorption:** The inability of the intestines to absorb certain nutrients properly.
- **Fat-soluble vitamins:** Vitamins A, D, E, and K that require fat to be absorbed and used by the body.
- **Bleeding diathesis:** A tendency to bleed easily, often due to vitamin K deficiency.
- **Osteomalacia:** Softening of bones due to vitamin D deficiency.
- **Alkaline phosphatase (ALP):** An enzyme found in the liver and bones; elevated levels are often a sign of liver or bile duct obstruction.
- **Aminotransferases (ALT/AST):** Enzymes found in the liver, elevated when liver cells are damaged.

## Ascites

- **Ascites:** Accumulation of excess fluid in the peritoneal cavity (the space within the abdomen).
- **Peritoneal cavity:** The space inside the abdomen that holds abdominal organs like the stomach, liver, and intestines.

## Aetiology (Causes)

- **Transudates:** Fluid accumulation in the peritoneum due to non-inflammatory causes, typically from systemic conditions like cirrhosis or heart failure.
- **Exudates:** Fluid accumulation due to inflammation or malignancy in the peritoneum.
- **Cirrhosis with portal hypertension:** Liver disease causing high pressure in the veins of the liver, leading to fluid accumulation in the abdomen.



- **Congestive heart failure (CHF):** A condition where the heart cannot pump blood efficiently, leading to fluid buildup.
- **Hypoproteinaemia:** Low protein levels in the blood, commonly seen in nephrotic syndrome or malnutrition.
- **Pancreatitis:** Inflammation of the pancreas that can lead to fluid buildup.
- **Biliary ascites:** Fluid accumulation due to bile leakage from the bile ducts.
- **Tuberculous peritonitis:** Inflammation of the peritoneum caused by tuberculosis infection.
- **Bacterial peritonitis:** Infection of the peritoneum caused by bacteria.
- **Malignancy (peritoneal, hepatic):** Cancer affecting the peritoneum or liver can cause fluid buildup.
- **Chylous ascites:** Fluid that contains lymph, often due to filariasis, trauma, or tumors.
- **Meig's syndrome:** A condition involving ascites, right-sided pleural effusion, and ovarian tumors.
- **Budd-Chiari syndrome:** A condition where veins in the liver are obstructed, leading to fluid buildup.

## Mechanisms of Formation

- **Increased capillary permeability:** Inflammation causes blood vessels to leak fluid into the peritoneum, as seen in peritonitis.
- **Venous obstruction:** Blockage of veins (e.g., inferior vena cava obstruction) causes fluid to leak into the abdominal cavity.
- **Lymphatic obstruction:** Blockage of lymphatic vessels leads to chylous ascites (fluid containing lymph).
- **Rupture of viscus:** A rupture of an abdominal organ (e.g., pancreas) can cause leakage of fluids into the peritoneal cavity.

## Clinical Features

- **Abdominal distension:** Bloating or swelling of the abdomen due to fluid accumulation.
- **Dyspnoea:** Shortness of breath.
- **Orthopnoea:** Difficulty breathing while lying down, commonly seen with ascites.
- **Indigestion:** Difficulty in digestion, often associated with abdominal pressure from ascites.
- **Heartburns:** A burning sensation in the chest due to reflux of stomach acid.
- **Fluid thrill:** A physical sign used to detect massive ascites by palpating the abdomen.
- **Shifting dullness:** A physical exam finding used to diagnose moderate ascites.
- **Spider nevi:** Small, spider-like blood vessels visible under the skin, common in chronic liver disease.



- **Palmar erythema:** Redness of the palms, often associated with liver disease.
- **Gynaecomastia:** Enlargement of male breast tissue.
- **Splenomegaly:** Enlargement of the spleen, often due to portal hypertension.
- **Resonance in epigastrium:** A sound heard during percussion of the upper abdomen, indicating floating intestines.

## Secondary Effects of Ascites

- **Scrotal oedema:** Swelling of the scrotum, often due to fluid buildup.
- **Pleural effusion:** Fluid accumulation in the pleural cavity (around the lungs), especially on the right side.
- **Oedema:** Swelling caused by excess fluid in the tissues.
- **Cardiac apex shift:** Displacement of the heart's tip due to the raised diaphragm from ascites.
- **Neck vein distension:** Swelling of the veins in the neck due to increased pressure in the heart.
- **Meralgia paraesthetica:** A condition caused by compression of the lateral cutaneous nerve, leading to tingling and numbness in the thigh.

## Investigations

- **Ultrasonography:** A diagnostic imaging technique that uses sound waves to detect fluid and assess causes.
- **Diagnostic paracentesis:** A procedure where a needle is used to remove fluid from the abdomen for examination.
- **Laparoscopy:** A minimally invasive surgical procedure to view the inside of the abdomen and collect tissue samples.
- **Ascitic fluid examination:** Analyzing the fluid taken from the abdomen to determine the cause of ascites.

## Management

- **Salt restriction:** Reducing salt intake to help manage fluid retention in ascites.
- **Diuretics:** Medications used to remove excess fluid from the body.
  - **Spironolactone:** A diuretic used to manage ascites, starting with 100 mg/day and increased as needed.
  - **Furosemide:** Another diuretic that may be used alongside spironolactone, typically in doses of 40-160 mg/day.
- **Therapeutic paracentesis:** A procedure to remove large amounts of fluid (3-5 liters) from the abdomen to relieve symptoms.
- **Surgical options:**
  - **Portacaval shunt:** A surgical procedure to bypass the liver in cases of refractory ascites.



- **TIPSS:** A procedure to place a stent in the liver to manage portal hypertension and ascites.